

Assignment 4: Tool Specification

Trace — Personal Productivity and Task Execution Agent

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The specification below documents all ten tools implemented in Trace: the eight required tools and two bonus tools (Create Reminder, Draft Follow-up Email). Each entry defines the tool's purpose, its exact input and output schema, which fields are required versus optional, whether it reads or writes data, whether human approval is required before it runs, the errors it can raise, and one worked example — enough for another developer to implement the tool correctly without seeing the source code.

Tool 1 — Create Task

Purpose	Creates a new task record so the user or the agent can track a piece of work going forward.
Input schema	<pre>{ "title": string, "description": string, "priority": "Low" "Medium" "High" "Critical", "due_date": string (ISO 8601 date, e.g. "2026-07-28"), "tags": [string] }</pre>
Required fields	title, priority
Optional fields	description, due_date, tags
Output schema	<pre>{ "task_id": string, "title": string, "status": "Pending", "created_at": string (ISO 8601 datetime), "confirmation": "created" }</pre>
Operation type	Write
Approval required	No for a single task created from a direct user request. Yes when creating multiple tasks in one action (e.g. the Meeting Notes to Tasks workflow), per the approval rule for batch task creation.
Possible errors	invalid_arguments (missing title or priority, priority not in the allowed set, due_date not a valid date) • storage_error (database write failed)
Example call	<pre>{ "tool": "create_task", "arguments": { "title": "Sync roadmap with engineering", "description": "Align Q3 roadmap with eng capacity", "priority": "Medium", "due_date": "2026-07-28", "tags": ["Roadmap"] } }</pre>
Example result	<pre>{ "task_id": "TSK-1042", "title": "Sync roadmap with engineering", "status": "Pending", "created_at": "2026-07-19T10:05:00Z", "confirmation": "created" }</pre>

Tool 2 — List Tasks

Purpose	Retrieves tasks that match optional filters, so the user or the agent can see what is outstanding.
Input schema	<pre>{ "status": "Pending" "In Progress" "Blocked" "Completed" "Cancelled" , "priority": "Low" "Medium" "High" "Critical", "due_before": string (ISO date), "due_after": string (ISO date), "tag": string }</pre>
Required fields	None. An empty filter set returns every task.
Optional fields	status, priority, due_before, due_after, tag
Output schema	<pre>{ "tasks": [{ "task_id": string, "title": string, "priority": string, "status": string, "due_date": string, "tags": [string] }], "total_count": integer }</pre>
Operation type	Read
Approval required	No
Possible errors	invalid_arguments (unknown status/priority value, due_after later than due_before) • storage_error (database read failed)
Example call	<pre>{ "tool": "list_tasks", "arguments": { "priority": "Critical", "due_before": "2026-07-26" } }</pre>

Example result

```
{
  "tasks": [
    {
      "task_id": "TSK-1035",
      "title": "Draft Q3 product launch plan",
      "priority": "Critical",
      "status": "In Progress",
      "due_date": "2026-07-21",
      "tags": ["Launch", "Strategy"]
    }
  ],
  "total_count": 1
}
```

Tool 3 — Update Task

Purpose	Modifies one or more fields of an existing task. This is a write action.
Input schema	<pre>{ "task_id": string, "title": string, "description": string, "priority": "Low" "Medium" "High" "Critical", "due_date": string (ISO date), "status": "Pending" "In Progress" "Blocked" "Completed" "Cancelled" , "tags": [string] }</pre>
Required fields	task_id, and at least one field to change
Optional fields	title, description, priority, due_date, status, tags
Output schema	<pre>{ "task_id": string, "updated_fields": object, "updated_at": string (ISO 8601 datetime) }</pre>
Operation type	Write
Approval required	Yes, always — must be confirmed by the user before execution.
Possible errors	not_found (unknown task_id) • invalid_arguments (no fields supplied, or an invalid enum value for priority/status) • storage_error
Example call	<pre>{ "tool": "update_task", "arguments": { "task_id": "TSK-1035", "status": "In Progress", "due_date": "2026-07-23" } }</pre>
Example result	<pre>{ "task_id": "TSK-1035", "updated_fields": { "status": "In Progress", "due_date": "2026-07-23" }, "updated_at": "2026-07-19T10:12:00Z" }</pre>

Tool 4 — Complete Task

Purpose	Marks an existing task as Completed and records when it was finished.
Input schema	<pre>{ "task_id": string }</pre>
Required fields	task_id
Optional fields	None
Output schema	<pre>{ "task_id": string, "status": "Completed", "completed_at": string (ISO 8601 datetime) }</pre>
Operation type	Write
Approval required	Yes, always — must be confirmed by the user before execution.
Possible errors	not_found (unknown task_id) • invalid_state (task is already Completed or Cancelled) • storage_error
Example call	<pre>{ "tool": "complete_task", "arguments": { "task_id": "TSK-1021" } }</pre>
Example result	<pre>{ "task_id": "TSK-1021", "status": "Completed", "completed_at": "2026-07-19T10:15:00Z" }</pre>

Tool 5 — Search Notes

Purpose	Finds notes relevant to a query, optionally narrowed by category or date range, ranked by relevance.
Input schema	<pre>{ "query": string, "category": string, "date_from": string (ISO date), "date_to": string (ISO date) }</pre>
Required fields	query
Optional fields	category, date_from, date_to
Output schema	<pre>{ "notes": [{ "note_id": string, "title": string, "category": string, "tags": [string], "updated_at": string, "snippet": string, "score": number }], "total_count": integer }</pre>
Operation type	Read
Approval required	No
Possible errors	invalid_arguments (empty query, or date_from later than date_to) • storage_error
Example call	<pre>{ "tool": "search_notes", "arguments": { "query": "onboarding flow feedback", "category": "Product" } }</pre>

Example result

```
{
  "notes": [
    {
      "note_id": "NTE-208",
      "title": "Onboarding usability session notes",
      "category": "Product",
      "tags": ["Onboarding", "UX"],
      "updated_at": "2026-07-15T09:00:00Z",
      "snippet": "Three of five users got stuck on step 2...",
      "score": 0.91
    }
  ],
  "total_count": 1
}
```


Tool 6 — Save Note

Purpose	Creates a new persistent note.
Input schema	<pre>{ "title": string, "content": string, "category": string, "tags": [string] }</pre>
Required fields	title, content
Optional fields	category, tags
Output schema	<pre>{ "note_id": string, "confirmation": "saved", "created_at": string (ISO 8601 datetime) }</pre>
Operation type	Write
Approval required	No — saving a single note is not in the required-approval list, but the action is still recorded in the execution log.
Possible errors	invalid_arguments (missing title or content) • storage_error
Example call	<pre>{ "tool": "save_note", "arguments": { "title": "Onboarding usability session notes", "content": "Three of five users got stuck on step 2 of setup...", "category": "Product", "tags": ["Onboarding", "UX"] } }</pre>
Example result	<pre>{ "note_id": "NTE-208", "confirmation": "saved", "created_at": "2026-07-19T10:20:00Z" }</pre>

Tool 7 — Extract Meeting Actions

Purpose	Parses meeting notes or a transcript into a structured summary of decisions, action items, and open questions.
Input schema	<pre>{ "transcript": string, "attendees": [string] }</pre>
Required fields	transcript
Optional fields	attendees (helps the model match action items to an owner)
Output schema	<pre>{ "summary": string, "decisions": [string], "action_items": [{ "description": string, "owner": string null, "deadline": string null }], "unresolved_questions": [string] }</pre>
Operation type	Read (analysis only — this tool does not write anything to the database by itself)
Approval required	No for the extraction step. The task(s) later created from its output require approval through Tool 1 / the Meeting Notes to Tasks workflow.
Possible errors	invalid_arguments (empty transcript, or transcript exceeds the model's context limit) • llm_error (model failed to return a valid structured response)
Example call	<pre>{ "tool": "extract_meeting_actions", "arguments": { "transcript": "We agreed to delay launch by one week. Ahmed will update the timeline...", "attendees": ["Ahmed", "Sara"] } }</pre>
Example result	<pre>{ "summary": "Team agreed to delay the launch by one week to finish QA.", "decisions": ["Delay launch by one week"], "action_items": [{ "description": "Update the project timeline", "owner": "Ahmed", "deadline": "2026-07-22" }], "unresolved_questions": ["Who will notify the marketing team of the new date?"] }</pre>

Tool 8 — Generate Work Plan

Purpose	Builds an ordered schedule for a given day from the user's open tasks, available time, and stated priorities.
Input schema	<pre>{ "available_hours": number, "date": string (ISO date), "task_ids": [string], "priorities": [string] }</pre>
Required fields	available_hours, date
Optional fields	task_ids (defaults to all open tasks if omitted), priorities
Output schema	<pre>{ "schedule": [{ "task_id": string, "title": string, "recommended_start": string, "estimated_hours": number }], "focus_areas": [string], "deferred_tasks": [{ "task_id": string, "reason": string }], "risk_warnings": [string] }</pre>
Operation type	Read (analysis / recommendation only — does not modify any task)
Approval required	No
Possible errors	invalid_arguments (available_hours is zero or negative, date is invalid) • llm_error
Example call	<pre>{ "tool": "generate_work_plan", "arguments": { "available_hours": 6, "date": "2026-07-20", "priorities": ["Q3 launch"] } }</pre>

Example result

```
{
  "schedule": [
    {
      "task_id": "TSK-1035",
      "title": "Draft Q3 product launch plan",
      "recommended_start": "09:00",
      "estimated_hours": 3
    }
  ],
  "focus_areas": ["Q3 launch"],
  "deferred_tasks": [
    { "task_id": "TSK-1040", "reason": "Lower priority, due date is 8 days out" }
  ],
  "risk_warnings": ["TSK-1035 is due tomorrow and only partially scheduled to day"]
}
```

Bonus Tool 1 — Create Reminder

Purpose	Schedules a reminder message for a specific time, optionally linked to a task.
Input schema	<pre>{ "message": string, "remind_at": string (ISO 8601 datetime), "task_id": string }</pre>
Required fields	message, remind_at
Optional fields	task_id
Output schema	<pre>{ "reminder_id": string, "status": "scheduled", "remind_at": string }</pre>
Operation type	Write
Approval required	Yes, always — creating a reminder is explicitly listed as requiring human approval.
Possible errors	invalid_arguments (remind_at is in the past, message is empty) • not_found (task_id was supplied but does not exist) • storage_error
Example call	<pre>{ "tool": "create_reminder", "arguments": { "message": "Follow up on investor deck feedback", "remind_at": "2026-07-24T09:00:00Z", "task_id": "TSK-1038" } }</pre>
Example result	<pre>{ "reminder_id": "RMD-77", "status": "scheduled", "remind_at": "2026-07-24T09:00:00Z" }</pre>

Bonus Tool 2 — Draft Follow-up Email

Purpose	Drafts a follow-up email based on a task, meeting, or pending item, for the user to review and choose whether to send.
Input schema	<pre>{ "recipient_context": string, "purpose": string, "key_points": [string], "tone": string }</pre>
Required fields	recipient_context, purpose
Optional fields	key_points, tone (e.g. "formal", "friendly"; defaults to a neutral professional tone)
Output schema	<pre>{ "subject": string, "body": string, "status": "draft" }</pre>
Operation type	Read / compose (produces a draft only; never sends or simulates sending anything by itself)
Approval required	Yes — required before the draft is treated as sent or simulated as sent. The draft is returned for the user's review, edit, and explicit approval first, matching the rule that sending or simulating an email needs approval.
Possible errors	invalid_arguments (missing recipient_context or purpose) • llm_error
Example call	<pre>{ "tool": "draft_followup_email", "arguments": { "recipient_context": "Investor who requested the updated deck", "purpose": "Send the revised investor deck and confirm the next call", "tone": "formal" } }</pre>
Example result	<pre>{ "subject": "Updated Investor Deck and Next Steps", "body": "Hello, please find the revised deck attached. Could we confirm a time this week to walk through the updates together?", "status": "draft" }</pre>